PHYS 599: Physics of Living Systems

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Reaction–diffusion and patterning

Turing instability, positional precision, and scaling

Cells and nuclei can acquire spatial information by sensing diffusing molecules. We develop the reaction–diffusion equation, derive the conditions for Turing instability, and then examine positional precision in the fruit fly embryo and scaling in frog development. These examples connect the transport of molecules to spatial patterns of gene expression.

Fall 2026

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1 From molecular motion to a reaction–diffusion equation

Until now, we have focused on what individual cells and molecules can do. We now ask how spatially distributed molecules can organize the behavior of many cells. We will first study the mathematics without specifying a particular organism. The same equation will then let us ask three different questions: how a uniform chemical system becomes patterned, how a nucleus reads its position from a concentration gradient, and how a developing embryo adjusts its pattern to its size.

1.1 A concentration can depend on both position and time

Let $c(x,t)$ denote the number of molecules per unit volume near position x at time t. We describe variation along one direction and assume the cross-sectional area is constant. This is a one-dimensional spatial model, but c still has units of number per volume. A partial derivative $\partial c/\partial t$ asks how the concentration changes while holding position fixed; $\partial c/\partial x$ asks how it changes with position at a fixed time.

Define the flux $J(x,t)$ to be the net number of molecules crossing unit area per unit time in the positive x direction. A negative J means net motion in the negative x direction. For ordinary diffusion in a dilute medium, Fick's law is

$$J = -D\frac{\partial c}{\partial x}. \tag{1}$$

Here $D > 0$ is the diffusion coefficient, with units of length squared per time. Individual molecules move randomly in both directions. The minus sign describes their net transport down a concentration gradient.

Consider a thin slice of thickness Δx and cross-sectional area A. Its molecular count changes because molecules enter, leave, or are produced and removed locally. Write the net local production rate per unit volume as $R(c,x,t)$, which can be positive or negative. Conservation gives

$$A\Delta x\,\frac{\partial c}{\partial t} \simeq AJ(x,t) - AJ(x+\Delta x,t) + A\Delta x\,R. \tag{2}$$

Divide by $A\Delta x$ and take the limit $\Delta x \to 0$:

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} + R. \tag{3}$$

Combining this conservation law with Fick's law, for a constant D, yields

$$\boxed{\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2} + R(c,x,t).} \tag{4}$$

This is a **reaction–diffusion equation**. The diffusion term redistributes molecules; the reaction term changes their local number or chemical identity. For example, a production rate $s(x,t)$ and first-order degradation rate constant k give $R = s - kc$. The units of k are inverse time, whereas the units of s and R are concentration per time.

In three dimensions, the second derivative becomes the Laplacian, $\nabla^2 c = \partial_x^2 c + \partial_y^2 c + \partial_z^2 c$. If D varies with position, Fick's law instead gives $\partial_x(D\partial_x c)$; we cannot generally pull D outside the derivative. For two reacting species, each equation has its own diffusion term and a reaction rate that can depend on both concentrations.

1.2 Initial conditions, boundary conditions, and steady states

The differential equation is not the whole model. An initial condition specifies $c(x, 0)$. Boundary conditions specify what happens at the ends of the spatial domain. On an interval $0 \le x \le L$, examples are a fixed concentration, $c(0, t) = c_0$; a fixed incoming flux, $-D\partial_x c(0, t) = j_0$; and a reflecting end, $\partial_x c(L, t) = 0$. A reflecting end allows no diffusive flux through it. A periodic domain identifies its two ends and will be useful for studying Turing instability without imposing a concentration source at either end.

A **steady state** satisfies $\partial_t c = 0$. It can still have a spatial gradient and a continuous flux: production, diffusion, and removal can balance at each position. Finding a steady-state solution does not show that the system approaches it. To answer that question, we must determine whether small perturbations grow or decay.

1.3 Why diffusion alone smooths a spatial variation

On an infinite or periodic domain, consider a small sinusoidal variation around a constant concentration $\bar{c}$:

$$c(x, t) = \bar{c} + a(t)\cos(qx), \qquad |a(t)| \ll \bar{c}. \tag{5}$$

The wavenumber q corresponds to a wavelength $\ell = 2\pi/q$. Substitution into the diffusion equation, with $R = 0$, gives

$$\frac{da}{dt} = -Dq^2 a, \qquad a(t) = a(0)e^{-Dq^2 t}. \tag{6}$$

Every nonuniform mode decays. Shorter wavelengths decay faster because q is larger. The timescale for smoothing a feature of size ℓ is of order ℓ^2/D; for this particular sinusoid the decay time is $1/(Dq^2) = \ell^2/(4\pi^2 D)$.

We will soon encounter the surprising possibility that adding diffusion to a stable system of reacting chemicals makes a nonuniform mode grow. Each diffusion term still smooths its own concentration field, but the coupling between the fields changes the stability of the combined system.

2 Secretion, sensing, and spatial information

Cells often coordinate their behaviors by secreting molecules and sensing molecules secreted by other cells. A cell's location then affects the concentrations it encounters, and these concentrations can affect which genes it expresses. A molecule whose spatial distribution helps specify different developmental responses is called a **morphogen**.

There are two linked parts to this description. The reaction–diffusion equation describes how the concentration field is established. A response function describes how a cell or nucleus reads that field. For example, a threshold response can switch a gene on where a concentration exceeds a particular value. A smooth concentration gradient can therefore produce a boundary between distinct gene-expression domains.

The source need not be every cell, and sensing a signal does not require secreting that same signal. In our first biological example, nuclei share the cytoplasm of an early fruit fly embryo and read a Bicoid gradient generated from maternally supplied mRNA. In our second example, secreted proteins and extracellular interactions contribute to patterning in frog embryos. Before examining these examples, we study interacting chemicals and ask when spatial order can emerge from a nearly uniform state.

3 The Turing mechanism

3.1 Main idea of the Turing mechanism

The Turing mechanism for spatial-pattern formation describes a uniformly spread-out molecules self-organizing into spatial-patterns (usually periodic stripes and dots). Specifically, the Turing mechanism states a seemingly uniform sea of molecules can have a small ripple at some location (i.e, slightly higher concentrations of the molecules at some location) and that this ripple could grow over time, causing the entire sea of molecules to reorganize their concentration profiles to form highly ordered patterns, including stationary waves, stripes, and dots. Under these conditions, a uniform field of molecules is unstable. Turing had three surprising insights that enabled the Turing mechanism to have a lasting impact. First, at least two chemicals are necessary for the classical diffusion-driven instability of a stable uniform state, with ordinary diffusion, considered here. Secondly - and the one that is perhaps the most surprising - is that diffusion can actually be destabilizing factor that causes the system to be unstable. This latter point is counter-intuitive since diffusion has the role of homogenizing a field of chemicals, by smoothing out spatial variations in the chemical concentrations, which is the opposite of pattern formations. Finally, Turing's third major insight was that the diffusion-caused instability can induce growth of a structure (i.e., local inhomogeneity in chemical concentration) to grow at a particular wavelength, which can seed patterns such as stripes with regular spacings.

3.2 Linear stability of a uniform mixture

As we will see, these patterns arise in this two-chemical model when the diffusion constants differ sufficiently for the given reaction kinetics. To generate such instability, the diffusion constants of the molecules and the interactions between them must satisfy a condition in addition to stability of the reaction dynamics without diffusion. To derive this condition, we start with the reaction-diffusion equation used for describing this form of Turing mechanism. We will, as usual, use a reaction-diffusion equation with two types of molecules:

$$\frac{\partial u}{\partial t} = D_1 \nabla^2 u + R_1(u, v) \tag{7a}$$

$$\frac{\partial v}{\partial t} = D_2 \nabla^2 v + R_2(u, v) \tag{7b}$$

where u and v are concentrations of molecules, A and I, D_1 and D_2 are diffusion coefficients of A and I, and R_1 and R_2 are reaction rates for A and I (i.e., net creation rates for A and I). The form of the reaction-rate functions, R_i, depends on the specific chemical system under consideration. Some of the functional forms for R_i are given special names such as the **Gierer-Meinhardt model**, **Schnakenberg model**, and **Thomas model**.

To obtain the Turing mechanism, we assume that when the concentration is uniform - that is, $u(x, y) = U$ and $v(x, y) = V$, where U and V are the steady-state constant profiles) - then $R_1 = 0$ and $R_2 = 0$. That is,

$$0 = R_1(U, V) \tag{8a}$$

$$0 = R_2(U, V) \tag{8b}$$

We will require this uniform steady state to be stable under the reaction dynamics alone. This is essential: diffusion must cause an instability that was absent before spatial coupling was introduced.

But now suppose that there are small spatial inhomogeneities, δu and δv - that is, $u(x,y) = U + \delta u(x,y)$ and $v(x,y) = V + \delta v(x,y)$. Then we can Taylor expand R_1 and R_2 near their values about the uniform concentrations U and V, retaining only the terms linear in the perturbations:

$$R_1(u,v) \simeq r_{11}\delta u + r_{12}\delta v \tag{9a}$$
$$R_2(u,v) \simeq r_{21}\delta u + r_{22}\delta v, \tag{9b}$$

where r_{ij}'s are the following Jacobian elements of R_i:

$$r_{i1} \equiv \left.\frac{\partial R_i}{\partial u}\right|_{(U,V)} \tag{10a}$$
$$r_{i2} \equiv \left.\frac{\partial R_i}{\partial v}\right|_{(U,V)} \tag{10b}$$

The Jacobian is the matrix of these local slopes. For example, r_{12} tells us how the net production rate of A changes when we slightly increase v, while holding u fixed. Its sign therefore tells us whether that small change promotes or suppresses the production of A. All the derivatives are evaluated at the same steady state (U, V).

By defining matrices,

$$\vec{D} \equiv \begin{bmatrix} D_1 & 0 \\ 0 & D_2 \end{bmatrix}, \qquad \vec{r} \equiv \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}, \tag{11}$$

and

$$\vec{c} \equiv \begin{bmatrix} \delta u \\ \delta v \end{bmatrix} \tag{12}$$

we can rewrite the linearized reaction-diffusion equations for the perturbations in terms of matrix multiplications as

$$\frac{\partial \vec{c}}{\partial t} = \vec{D}\nabla^2\vec{c} + \vec{r}\vec{c} \tag{13}$$

To solve Eq. 13, we must first impose boundary conditions (and also state where our boundaries are). For this calculation, we can impose either a periodic boundary condition (thus still keeping the system finite) or assume an infinite space without boundaries. In either case, we can analyze spatial Fourier modes because the coefficients in Eq. 13 are uniform in space. The equations have translational symmetry; an individual patterned solution need not have that symmetry. We will confine ourselves one-dimension as a three-dimensional system with periodic or infinite boundaries proceed in the same way. We consider spatial perturbations of the form e^{iqx} and temporal perturbations of the form $e^{k_q t}$, where k_q is a complex number and q is a real number. Since we can represent more general perturbations in space by a Fourier series on a periodic domain, or a Fourier integral on an infinite domain, and Eq. 13 is linear, it is sufficient to analyze the individual Fourier modes. Specifically, we seek solutions of the form:

$$\vec{c} \equiv \begin{bmatrix} u_o \\ v_o \end{bmatrix} e^{iqx} e^{k_q t}, \tag{14}$$

where u_o and v_o are constant amplitudes of perturbation. The physical perturbation is the real part of this expression. A mode has wavelength $2\pi/|q|$, and $\partial_x^2 e^{iqx} = -q^2 e^{iqx}$: this is why diffusion contributes a negative term proportional to q^2. The mode $q = 0$ changes the concentrations uniformly everywhere.

On a periodic interval of length L, the allowed wavenumbers are $q_n = 2\pi n/L$, with integer n. For reflecting (no-flux) boundaries at $x = 0, L$, cosine modes instead have $q_n = n\pi/L$, with $n = 0, 1, 2, \ldots$. We can first find which wavenumbers would grow on an infinite domain and then check whether a finite domain admits any of them.

The real component of k_q (i.e., $\mathrm{Re}(k_q)$) is the rate of growth or shrinkage of the spatial perturbation. Specifically, if $\mathrm{Re}(k_q) < 0$, then this mode of the initial perturbation decays over time - the system is called "linearly stable" if all allowed modes decay because Eq. 9b used a linear approximation - whereas $\mathrm{Re}(k_q) > 0$ leads to the initial perturbation growing over time, at least for a time when the linear approximation (Eq. 9b) is valid. Thus the problem of finding if uniform concentrations of A and I are unstable reduces to determining the sign of the real part of each growth rate k_q for each q. Plugging the perturbation (Eq. 14) into the reaction-diffusion equation (Eq. 13), we obtain

$$k_q \vec{c} = (\vec{r} - \vec{D}q^2)\vec{c} \tag{15}$$

Then by defining $\vec{P}_q = \vec{r} - \vec{D}q^2$, we obtain the following eigenvalue-eigenvector equation:

$$k_q \vec{c} = \vec{P}_q \vec{c}, \tag{16}$$

where k_q is an eigenvalue and the amplitude vector $(u_o, v_o)^T$ is its eigenvector. An eigenvector specifies a ratio of the two perturbations that the matrix changes only by an overall factor; that factor is the eigenvalue. Here, that overall factor is the temporal growth rate. A 2×2 matrix generally gives two growth rates for each q, and both must have negative real parts for stability. Thus solving the reaction-diffusion equation with a small perturbation has reduced to finding the eigenvalue and an eigenvector for each q. For $\vec{c}$ to be non-zero, Eq. 16 tells us that

$$0 = det(\vec{P}_q - k_q \vec{I}) \tag{17a}$$

$$= k_q^2 - tr(\vec{P}_q)k_q + det(\vec{P}_q) \tag{17b}$$

where $\vec{I}$ is the 2 x 2 identity matrix. For a matrix $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$, its trace is $a + d$ and its determinant is $ad - bc$. Solving this quadratic equation, we obtain the two eigenvalues, k_q

$$k_q = \frac{1}{2} tr(\vec{P}_q) \pm \frac{1}{2}\sqrt{(tr(\vec{P}_q))^2 - 4det(\vec{P}_q)} \tag{18}$$

Eq. 18 is perhaps the most important equation related to the Turing mechanism because it tells us everything that we need to know about when a uniform concentration field becomes unstable and what the spatial length (defined by q) associated with the instability is. We can diagrammatically summarize our finding (Fig. 1):

Thus, for a uniformly mixed A and I (i.e., $u = U, v = V$) to be **stable**, we need *both* of the following conditions to be satisfied for every allowed q:

$$tr(\vec{P}_q) = r_{11} + r_{22} - (D_1 + D_2)q^2 < 0 \qquad \text{(Stability condition 1)} \tag{19a}$$

$$det(\vec{P}_q) = (r_{11} - D_1 q^2)(r_{22} - D_2 q^2) - r_{12} r_{21} > 0 \qquad \text{(Stability condition 2)} \tag{19b}$$

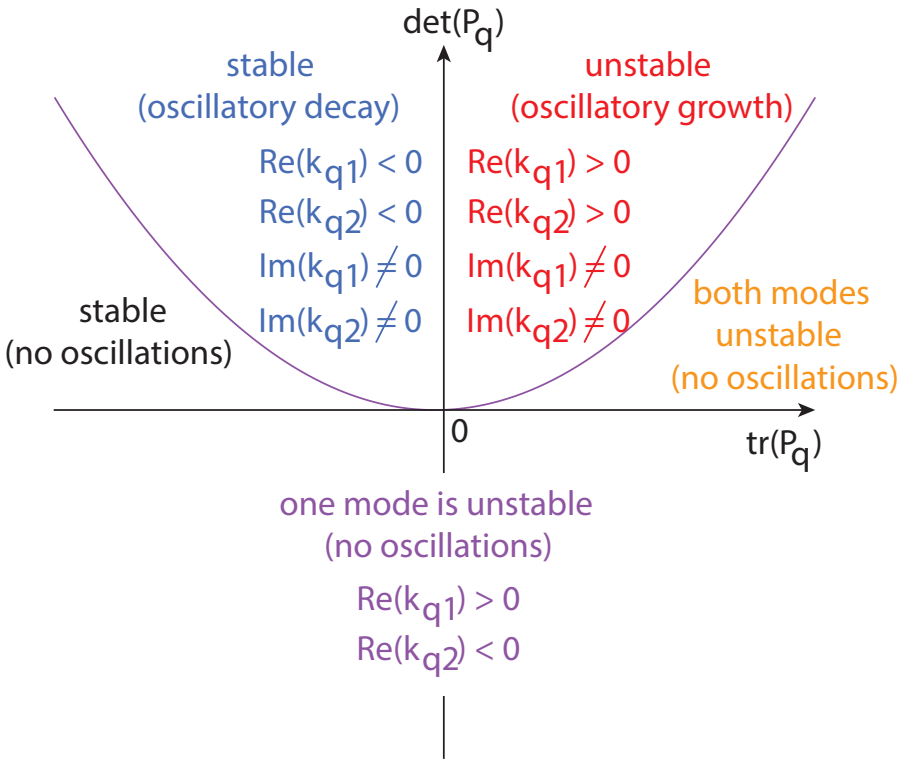


Figure 1: Graphical summary of linear stability (graphical representation of Eq. 18). The parabola represents $[tr(\vec{P}_q)]^2 = 4det(\vec{P}_q)$. For the two-species Turing instability considered here, the reaction dynamics starts in the stable left half-plane; increasing q can take the determinant below zero while the trace remains negative.

If the $q = 0$ mode is stable but an allowed non-zero wave number has a growth rate with positive real part, then we say that the chemical system exhibits a **Turing instability**; the onset of this instability is a **Turing bifurcation**. Equality at a stability boundary can give a marginal mode and does not by itself establish growth. If the system consists of N chemicals, then our P_q would be a N x N matrix and our problem would be to solve for N eigenvalues (if any one of them has positive real part at an allowed non-zero q, while every eigenvalue at $q = 0$ has negative real part, then diffusion has destabilized the uniform steady state). Turing's insight was that he focused on a two-chemical system, which simplified analysis while exhibiting non-trivial behaviours.

Turing's main physical insight was that diffusion could drive the instability. To see this, suppose we start with a uniform mixture of A and I by, for example, constantly stirring the beaker that contains both chemicals. Stirring does not set the molecular diffusion constants to zero. Instead, a uniform perturbation has $q = 0$, so the diffusion terms vanish and we test the reaction dynamics alone. Then Eqs. 19a and 19b become

$$tr(\vec{P}_0) = r_{11} + r_{22} < 0 \tag{20a}$$

$$det(\vec{P}_0) = r_{11}r_{22} - r_{12}r_{21} > 0 \tag{20b}$$

Eq. 20a and the fact that the diffusion constants and q^2 are non-negative mean that for $D_1, D_2 > 0$ and $q > 0$, we have

$$tr(\vec{P}_q) = r_{11} + r_{22} - (D_1 + D_2)q^2 \tag{21a}$$
$$< r_{11} + r_{22} \tag{21b}$$
$$< 0 \tag{21c}$$

Thus, Eq. 19a is automatically satisfied, when we include diffusion, if a well-mixed condition is stable under the reaction dynamics alone. Hence, the only way that the uniformly mixed chemicals become unstable is if $det(\vec{P}_q) < 0$ for some allowed non-zero q (i.e., if the system resides in the lower left quadrant in Fig. 1). Note that the determinant of $\vec{P}_q$ is quadratic in q^2 with a positive factor in front of it (Eq. 19b). This means that the graph of $\det(\vec{P}_q)$, as a function of q^2, is a parabola with its opening facing upwards (in positive direction). Thus we only need to know if the minimum of this parabola is below zero. If not, then there is no Turing instability. If there is and this minimum value occurs at a real non-zero q, then there is a band of unstable wavenumbers on an infinite domain. A finite domain must admit a mode within that band. The miminum value occurs at $q = q_o$, where

$$0 = \frac{d(det(\vec{P}_q))}{d(q^2)}\bigg|_{(q_o)^2}, \tag{22}$$

Solving above equation, we find

$$(q_o)^2 = \frac{D_1 r_{22} + D_2 r_{11}}{2D_1 D_2} \tag{23}$$

and that the minimum value of the determinant is

$$det(\vec{P}_{q_o}) = r_{11}r_{22} - r_{12}r_{21} - \frac{(D_1 r_{22} + D_2 r_{11})^2}{4D_1 D_2} \tag{24}$$

From Eq. 24, we see that the stability condition 2 (Eq. 19b) is violated (i.e., there exists a value of q for which $\det(\vec{P}_q) < 0$) when,

$$D_1 r_{22} + D_2 r_{11} > 2\sqrt{D_1 D_2 (r_{11}r_{22} - r_{12}r_{21})} \tag{25}$$

Given stable reaction dynamics, $D_1, D_2 > 0$, and a continuum of spatial wavenumbers, Eq. 25 is the additional condition that is necessary and sufficient for Turing instability. For a finite domain, an allowed mode must also lie in the unstable band.

To collect all the requirements in one place, define $\Delta = r_{11}r_{22} - r_{12}r_{21}$ and $S = D_1 r_{22} + D_2 r_{11}$. The four inequalities are

$$r_{11} + r_{22} < 0, \qquad \Delta > 0, \tag{26}$$
$$S > 0, \qquad S^2 > 4D_1 D_2 \Delta. \tag{27}$$

The first two make the uniform mode stable. The last two make $det(\vec{P}_q) = \Delta - Sq^2 + D_1 D_2 q^4$ negative over the band

$$q_-^2 < q^2 < q_+^2, \qquad q_\pm^2 = \frac{S \pm \sqrt{S^2 - 4D_1 D_2 \Delta}}{2D_1 D_2}. \tag{28}$$

The q_o above minimizes the determinant. The fastest-growing mode instead maximizes the larger growth rate in Eq. 18; these wavenumbers coincide at the continuum onset of instability, but need not coincide

away from onset. Linear analysis describes the initial growth. The nonlinear terms, neglected in the expansion, determine whether this growth saturates into a stable pattern and what that pattern looks like.

3.3 Local activation and long-range inhibition

Let us now analyze some consequences of Eq. 25. First, we note that Eq. 20b and Eq. 25 together imply $D_1 r_{22} + D_2 r_{11} > 0$. Since the diffusion coefficients are positive and $r_{11} + r_{22} < 0$, r_{11} and r_{22} must have opposite signs. Without loss of generality, let us say that $r_{11} > 0$ and $r_{22} < 0$. Then

$$D_1 r_{22} + D_2 r_{11} > 0 \tag{29a}$$

$$\implies \frac{D_1}{D_2} < \frac{-r_{11}}{r_{22}} \tag{29b}$$

Then A, with diffusion coefficient D_1, is called an **activator** and I, with diffusion coefficient D_2, is called an **inhibitor**. Here, $r_{11} > 0$ means that A locally promotes its own production, while $r_{22} < 0$ means that an increase in I suppresses its own net production. In the usual activator–inhibitor interpretation, $r_{12} < 0$ and $r_{21} > 0$: I suppresses A, and A promotes I. The derivation also allows the opposite cross-interaction signs, so the diagonal signs alone do not establish this particular biochemical interpretation.

A second consequence of Eq. 25 is that **the activator must have a lower diffusion coefficient than the inhibitor for the Turing instability**. Indeed, Eq. 29b gives $D_2/D_1 > -r_{22}/r_{11}$, and stability of the uniform mode gives $-r_{22}/r_{11} > 1$. Hence $D_2 > D_1$. To express the related difference in spatial ranges, we rewrite the stability condition (Eq. 25) in terms of the characteristic diffusion lengths of A and I, which are L_1 and L_2 respectively:

$$L_1 \equiv \sqrt{\frac{D_1}{r_{11}}} \qquad L_2 \equiv \sqrt{\frac{-D_2}{r_{22}}} \tag{30}$$

These lengths compare diffusion to the local self-activation and self-relaxation rates. In particular, L_1 is not a decay length of an isolated stable species, because $r_{11} > 0$.

In terms of L_1 and L_2, the instability condition (Eq. 25) becomes

$$\frac{1}{2}\left(\frac{1}{L_1^2} - \frac{1}{L_2^2}\right) > \sqrt{\frac{r_{11}r_{22} - r_{12}r_{21}}{D_1 D_2}} > 0 \tag{31}$$

Thus it follows that for a uniform mixture of chemicals to be unstable to diffusion, a necessary but not sufficient condition is that the activator has a shorter characteristic diffusion length than the inhibitor (i.e. $L_1 < L_2$). This alone is insufficient for having instability because the right side of the inequality in Eq. 31 must still be satisfied - this is not guaranteed by having $L_1 < L_2$. The condition that $L_1 < L_2$ is why we often say "**local activation and long-range inhibition**" for Turing instability. Finding real chemical systems that satisfy Eq. 31 has been difficult. For one, Eq. 29b sets a lower bound on how much larger the inhibitor's diffusion constant must be. For example, if the ratio $-r_{22}/r_{11}$ is 10, then D_2/D_1 must exceed 10 even before the stronger instability condition is checked. This is a large number.

3.4 Equal diffusion coefficients and a concrete example

There is a particularly simple way to see why unequal diffusion coefficients are necessary here. If $D_1 = D_2 = D$, then $\vec{P}_q = \vec{r} - Dq^2\vec{I}$. Its eigenvalues are therefore

$$k_{q,\pm} = k_{0,\pm} - Dq^2. \tag{32}$$

If the reaction dynamics is stable, $\mathrm{Re}\, k_{0,\pm} < 0$, subtracting Dq^2 can only make the perturbations decay more rapidly. Equal diffusion coefficients cannot generate the instability. This argument concerns the ordinary diagonal diffusion terms used here.

As a concrete example, choose units of time and length so that the following quantities are dimensionless:

$$\vec{r} = \begin{pmatrix} 1 & -2 \\ 2 & -3 \end{pmatrix}, \qquad D_1 = 1, \qquad D_2 = 20. \tag{33}$$

The uniform mode has $tr(\vec{r}) = -2$ and $det(\vec{r}) = 1$, with both eigenvalues equal to -1. Thus the reaction dynamics eventually returns small perturbations toward the uniform steady state. With diffusion,

$$tr(\vec{P}_q) = -2 - 21q^2, \qquad det(\vec{P}_q) = 1 - 17q^2 + 20q^4. \tag{34}$$

The trace stays negative, but the determinant is negative when

$$\begin{aligned} \frac{17 - \sqrt{209}}{40} < q^2 < \frac{17 + \sqrt{209}}{40}, \\ \text{or approximately} \qquad 0.252 < q < 0.887 \quad (q > 0). \end{aligned} \tag{35}$$

These nonuniform modes grow. The uniform mode and the very short-wavelength modes decay (Fig. 2). On a periodic interval with $L = 20$, the modes $q_1 = 2\pi/20$ and $q_2 = 4\pi/20$ lie in the unstable band. On an interval with $L = 5$, even the smallest non-zero wavenumber is too large: this finite system has no Turing instability despite satisfying the four continuum inequalities. The boundary conditions and system size are therefore part of the physical problem.

3.5 From a mathematical mechanism to biological patterning

Turing's paper considered both idealized cells and continuous tissue [1]; here we have studied the chemical continuum formulation without explicit cell dynamics. Experiments support Turing-type mechanisms in several developmental systems, including mouse palatal ridges and digit specification [2, 3]. However, stripes or spots by themselves do not demonstrate a diffusion-driven instability, and a biological system need not implement exactly this two-species model. We now turn to two examples in which diffusing morphogens organize spatial patterns: a gradient read by nuclei in the fly embryo, and extracellular signaling that adjusts a pattern to the size of the frog embryo.

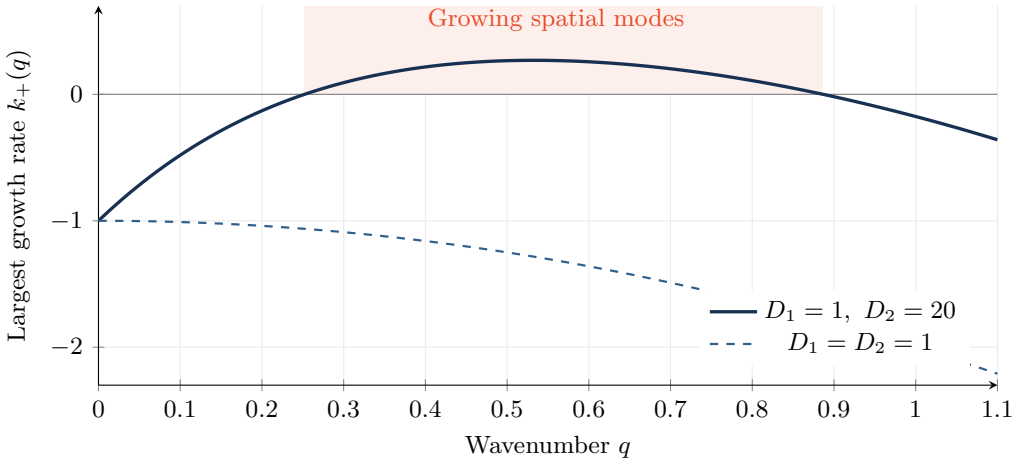


Figure 2: Largest growth rate for the dimensionless example in Eq. 33. The solid curve becomes positive only over a finite band of non-zero wavenumbers. The dashed curve shows the same reaction matrix with equal diffusion coefficients, for which every mode decays. Linear instability identifies which small perturbations initially grow; this example specifies the local reaction derivatives, not a full nonlinear model for their eventual saturation.

4 Applying the Berg-Purcell limit to the fruit fly embryo

In this section, we briefly summarize beautiful, physics-type high-precision experiments that Thomas Gregor and colleagues report in:

- T. Gregor, D. W. Tank, E. F. Wieschaus, and W. Bialek. Probing the limits to positional information, *Cell* (2007) [5]

4.1 A maternal source and a field of nuclei

In this paper, Gregor et al. investigated how accurately nuclei inside the fruit fly embryo can measure the concentration of the morphogen called *Bicoid* that surrounds them. At the cortical blastoderm stages of the early fruit fly embryo, individual nuclei, not cells, are arranged next to each other in a near triangular lattice. The nuclei have their own nuclear envelopes during interphase, but they are not yet separated into individual cells by plasma membranes; they share a common cytoplasm containing other organelles. Such an embryo is called a **syncytium**. Simply put, the mother deposits the Bicoid mRNA at one end of the embryo. This end will form the head (i.e., *anterior*). Ribosomes that diffuse inside the embryo encounter and translate the mRNA into the Bicoid proteins. The Bicoid proteins move diffusively in the embryo, with a net flux away from the anterior source towards the other end, which will form the tail (i.e., *posterior*). In so doing, a Bicoid-concentration gradient forms. Each nucleus in the embryo must "measure" the concentration of the Bicoid around itself. The concentration of the Bicoid that a nucleus measures is a function of the position of the nucleus relative to the embryo's head, where the Bicoid concentration would be the highest (since that is where the translation of Bicoid occurs - the mRNA is essentially localized at the embryo's head). Thus a nucleus can potentially "know" how far from the head it is by accurately measuring the Bicoid concentration. The arrival of diffusing Bicoid molecules at their binding sites limits the precision of a concentration measurement made with specified sensing resources, as in the Berg-Purcell argument. Gregor et al. quantified how accurate is ?accurate enough? with calculations and tour de force, physics-type precision experiments on the fly embryo. For their

experiments, Gregor et al. used embryos expressing Bicoid fused to Green Fluorescent Protein (GFP), and calibrated the fluorescence against a GFP solution of known concentration to estimate absolute Bicoid concentrations. This measured a concentration from fluorescence intensity, rather than resolving and counting individual Bicoid proteins.

The mother therefore supplies the localized mRNA source; she does not deposit a ready-made Bicoid protein gradient. The gradient is generated in the embryo by protein production and transport, together with loss and exchange processes. This is a spatially coordinated response to a maternally supplied signal. It does not require the nuclei to secrete Bicoid to one another.

4.2 A reaction–diffusion model for the gradient

We now discuss the paper's main findings. Two hours after the fruit fly egg has been fertilized, the length L of the embryo is approximately 500 μm and the average distance Δx between the centers of two adjacent nuclei (which we treat as spheres) that are aligned along the straight line joining the head to tail (anterior-posterior) is approximately 8 μm (see Figure in the paper). Each nucleus has a diameter of approximately 5 μm. Based on these measurements, we have

$$\frac{\Delta x}{L} = \frac{8\ \mu\text{m}}{500\ \mu\text{m}} \approx 0.016 \tag{36}$$

Thus having two adjacent nuclei being able to tell that they are at two different locations along the anterior-posterior axis means that the nuclei must be able to resolve a difference in length by about 1.6% of the embryo's total length. Suppose that a nucleus is at position x where the Bicoid concentration is $c(x)$. Let $\Delta c + c(x)$ be the Bicoid concentration on its adjacent nucleus. To connect this observation to the reaction–diffusion equation, let $c(x,t)$ represent an effective Bicoid concentration along the anterior–posterior axis. A simple synthesis–diffusion–degradation model is

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2} - kc + s(x), \tag{37}$$

where $k > 0$ is a first-order loss rate and $s(x)$ is a localized source. This is a coarse-grained model: it does not separately track cytoplasmic and nuclear Bicoid or the changes during nuclear division. Its usefulness here is that it shows how transport and loss can set a spatial length.

We can idealize the narrow anterior production region as an imposed incoming flux j_0 at $x = 0$. Away from that source, $s = 0$. At steady-state,

$$D\frac{d^2 c_s}{dx^2} - kc_s = 0, \qquad c_s(x) = Ae^{x/\lambda} + Be^{-x/\lambda}, \qquad \lambda = \sqrt{\frac{D}{k}}. \tag{38}$$

The quantity λ has units of length: in a lifetime of order k^{-1}, a molecule diffuses a distance of order $\sqrt{D/k}$. For a semi-infinite domain, the condition $c_s(x) \to 0$ as $x \to \infty$ sets $A = 0$. The source condition is

$$-D\left.\frac{dc_s}{dx}\right|_{x=0} = j_0, \qquad B = \frac{j_0\lambda}{D} \equiv c_0. \tag{39}$$

Thus the steady-state profile, which we now write as $c(x)$, is

$$c(x) = c_0 e^{-x/\lambda} \tag{40}$$

where λ is the diffusion length. Diffusion alone, without loss or an appropriate sink, does not give this steady exponential profile. For example, with fixed incoming flux, a closed finite domain and no degradation, the total amount keeps increasing.

A real embryo is finite. If we use $c_s'(L) = 0$ at a reflecting posterior boundary, the exact steady solution of the same model is

$$c_s(x) = \frac{j_0 \lambda}{D} \frac{\cosh[(L-x)/\lambda]}{\sinh(L/\lambda)}. \tag{41}$$

It approaches the exponential away from the posterior boundary when L/λ is large. Thus the exponential is a useful approximation with stated boundary assumptions, not a consequence of a source and diffusion alone. An approximately exponential measured profile also does not by itself establish this entire microscopic model; nuclear exchange and developmental dynamics must be considered separately [6].

Gregor et. al. have measured the concentration of the Bicoid inside tens of fly embryos. They found an approximately exponential decline in concentration from anterior to posterior. Intriguingly, they found that there was very little variability in the concentration profile $c(x)$ from one embryo to another, despite coming from different mothers. In a later study, Gregor's lab found that the number of maternally supplied *bicoid* mRNAs was reproducible to within about 9% between embryos; the result was precise provisioning, not an identical integer count in every embryo [7].

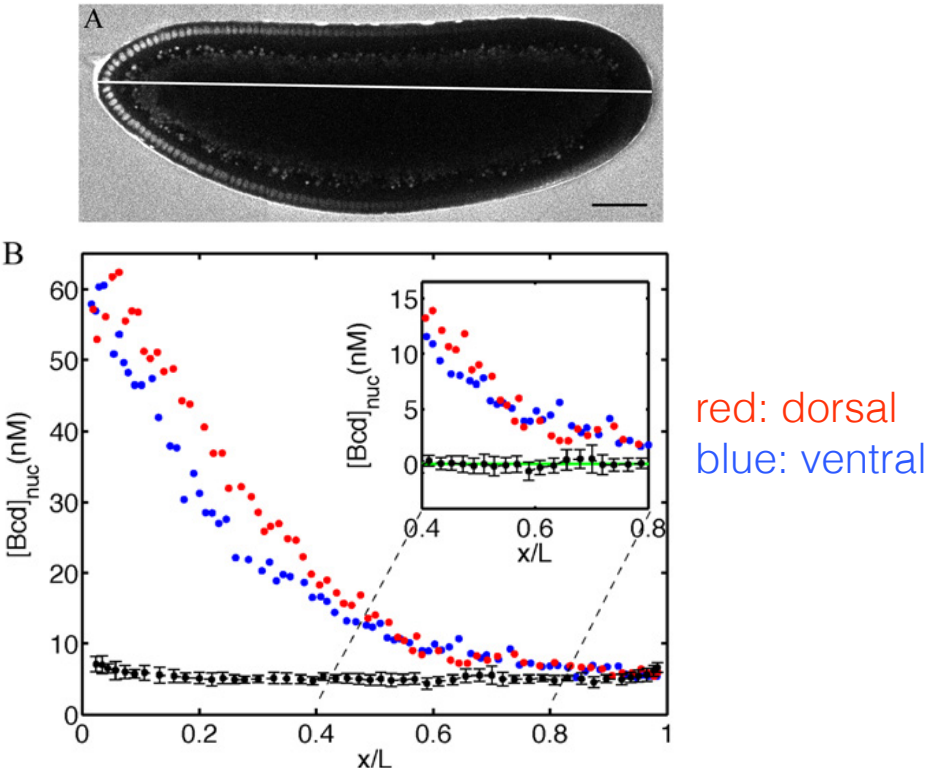


Figure 3: Bicoid concentration in the fly embryo, reproduced from the earlier course slides using Fig. 2 of Gregor et al. [5]. (A) Nuclear Bicoid–GFP fluorescence in a living embryo; the scale bar is 50 μm. (B) Apparent nuclear concentrations along the anterior–posterior axis; red and blue denote dorsal and ventral positions. Black points estimate fluorescence background using embryos without GFP. The inset shows background-subtracted concentrations near mid-embryo. The measured intensity is calibrated to a concentration.

4.3 How accurately must a nucleus distinguish concentration?

Their measurements showed that the diffusion length λ is approximtaely 100 μm. For two adjacent nuclei, we have

$$\left|\frac{\Delta c}{c(x)}\right| \simeq \left|\frac{1}{c(x)}\frac{dc}{dx}\right|\Delta x \tag{42a}$$

$$= \frac{\Delta x}{\lambda} \tag{42b}$$

$$= \frac{8\ \mu\text{m}}{100\ \mu\text{m}} \tag{42c}$$

$$= 0.08 \quad \text{(of order 10\%)}. \tag{42d}$$

According to Eq. 42d, the Bicoid concentration on a given nucleus is different by approximately 10% from the Bicoid conentration on its adjacent nucleus. Thus for two adjacent nuclei to "know" that they are apart by a distance of Δx, they must be able to distinguish 10% or less of a difference in their respective Bicoid concentrations.

The derivative approximation assumes $\Delta x \ll \lambda$. For comparison, the exact fractional decrease from x to $x + \Delta x$ in the exponential profile is $1 - e^{-\Delta x/\lambda}$, which is about 0.077 for these numbers.

The same calculation converts concentration noise into positional uncertainty. Suppose the mean profile is monotonic and a nucleus infers its position from a measured concentration. For a small concentration error δc, Taylor expansion gives

$$\delta c \simeq \frac{dc}{dx}\,\delta x. \tag{43}$$

Taking standard deviations,

$$\sigma_x \simeq \frac{\sigma_c}{|dc/dx|} = \lambda\frac{\sigma_c}{c} \qquad \text{(exponential profile)}. \tag{44}$$

A 10% concentration uncertainty with $\lambda = 100\,\mu$m therefore corresponds to a positional uncertainty of about $10\,\mu$m. This calculation concerns the information available from this one concentration, with the profile assumed known. It does not say that all developmental decisions use Bicoid alone.

4.4 Applying the sensing estimate from lecture 1

In lecture 1, we derived the Berg–Purcell sensing form for a concentration measurement averaged over a time T. Here, we use a for the detector size, so that L can continue to denote the embryo length:

$$\frac{\sigma_c}{c} \sim \frac{\alpha}{\sqrt{cDaT}}. \tag{45}$$

Here α is a dimensionless order-one constant. The prefactor depends on the detector, its binding dynamics, and the estimator. This is a characteristic precision scale under specified assumptions; calling it a rigorous lower bound requires specifying those assumptions and the available sensing resources [4].

The crucial issue is what counts as a detector. A nucleus is several micrometers across, but a regulatory binding site does not instantaneously count every Bicoid molecule inside that nucleus. It responds to molecules arriving at the site. In Gregor et al.'s estimate, the detector was therefore a DNA binding site with a linear size of order $a = 3\,$nm.

Near the Hunchback expression boundary, the measured Bicoid concentration was about 8 nM. Since 1 nM corresponds to approximately 0.602 molecules per μm^3, this gives

$$c \simeq 4.8\ \text{molecules}/\mu\text{m}^3. \tag{46}$$

Using the historical estimate $D \simeq 1\,\mu\text{m}^2/\text{s}$ and $a = 0.003\,\mu\text{m}$, and taking the order-one prefactor as one, gives

$$cDa \simeq (4.8)(1)(0.003)\,\text{s}^{-1} = 0.0144\,\text{s}^{-1}, \tag{47}$$

$$\frac{\sigma_c}{c} \sim \sqrt{\frac{69\,\text{s}}{T}}, \tag{48}$$

$$T \sim \frac{1}{cDa(0.1)^2} \simeq 6.9 \times 10^3\,\text{s} \simeq 1.9\,\text{h}. \tag{49}$$

This is the origin of the paper's roughly 7,000-second estimate for achieving 10% precision by local temporal averaging at a binding site. The developmental system changes substantially over such a time. The calculation therefore poses a puzzle about how the readout acquires sufficient information on the available timescale [5].

This estimate uses a particular effective diffusivity and sensing geometry. It is not a universal two-hour requirement for every conceivable Bicoid sensing mechanism. Its dependence on the assumed resources is precisely why the calculation is useful: it tells us which additional processes could matter. For example, multiple effectively independent measurements can reduce the noise, whereas repeated measurements of the same fluctuating signal cannot be counted as independent.

4.5 The gene-expression readout: Bicoid and Hunchback

Bicoid is a transcription factor: it affects the expression of downstream genes, including *hunchback*. Let H denote the Hunchback protein concentration and let $\overline{H}(c)$ be its mean value among nuclei with similar Bicoid concentration c. The measured mean input/output relation was well described by

$$\overline{H}(c) = H_{\max}\frac{c^n}{c^n + c_{1/2}^n}, \qquad n \simeq 5, \tag{50}$$

where $c_{1/2}$ is the Bicoid concentration at half-maximal response. The fitted Hill exponent describes the steepness of this relation; the fit alone does not establish that exactly five molecules bind simultaneously in a unique microscopic mechanism.

A steep response helps turn a graded input into distinct expression levels. But a steep response also amplifies input fluctuations. To compare the measured output noise with the sensing calculation, we convert it back into an equivalent input noise. For small fluctuations,

$$\delta H \simeq \frac{d\overline{H}}{dc}\,\delta c, \qquad \left(\frac{\sigma_c}{c}\right)_{\text{equiv}} = \frac{\sigma_H(c)}{c\,|d\overline{H}/dc|} = \frac{\sigma_H(c)}{|d\overline{H}/d\ln c|}. \tag{51}$$

For the Hill curve,

$$\frac{d\overline{H}}{d\ln c} = nH_{\max}\frac{c^n c_{1/2}^n}{(c^n + c_{1/2}^n)^2}, \qquad \left.\frac{d\overline{H}}{d\ln c}\right|_{c=c_{1/2}} = \frac{nH_{\max}}{4}. \tag{52}$$

This conversion matters because the same amount of output noise means different input precision on the steep and flat parts of the response curve. Near the transition, Gregor et al. inferred an equivalent input noise of order 10%, bringing the measured readout into the range required to distinguish neighboring positions.

4.6 Can averaging over space help?

The syncytial embryo offers a possible additional resource: molecules can move through the shared cytoplasm and thereby couple nearby nuclear responses. If a readout averages N independent local

estimates of equal variance, its standard deviation falls by $1/\sqrt{N}$. To see the limitation of that statement, write the mean of N estimates X_i as $\overline{X} = N^{-1}\sum_i X_i$. Then

$$\mathrm{Var}(\overline{X}) = \frac{1}{N^2}\left[\sum_i \mathrm{Var}(X_i) + 2\sum_{i<j}\mathrm{Cov}(X_i, X_j)\right]. \tag{53}$$

The familiar $\mathrm{Var}(X_i)/N$ result requires the covariances to vanish. Positive correlations reduce the gain in independent information. Spatial averaging also smooths a spatial signal, so increased precision can come at the cost of spatial resolution.

Gregor et al. proposed spatial averaging as a possible way to reconcile the observed precision with the time available, and measured correlations in Hunchback fluctuations across neighboring nuclei. Such correlations are consistent with spatial coupling, but do not by themselves establish a unique averaging mechanism. We should therefore keep the observation, the sensing estimate, and the proposed explanation separate.

The fly example has connected a concentration profile to gene expression and to positional precision. We now ask a different question: if the embryo is larger or smaller, how can a morphogen profile specify the same *relative* positions? A length $\lambda = \sqrt{D/k}$ set by fixed material parameters does not automatically grow with the embryo's length L.

5 Frog embryos: scaling of body parts in developing embryos

You and I have different heights and body sizes. Yet your and my arms are above our torsos and below our heads. Your and my legs are below our torsos. Given that you and I can be of vastly different heights, the my arms' lengths can largely deviate from those of your arms. But the length of my arms compared to the length of my height, in fact, would not be much different from those of yours. This also holds true for others animals and in fact, is crucial for them to properly function given that two animals from the same species will generally have different sizes, based on how they grew up and their genetic makeup. These facts, that the proportion of one body part relative to another body part in an organism is the conserved is called **scaling**. In this lecture, we will discuss an example of scaling in developing frog embryos and learn about numerical screening, a computational technique that one uses to screen for regulatory circuits that enable scaling in embryos (but it is also useful in other contexts).

5.1 The organizer and the biological problem

The fly embryo shows how a concentration gradient can provide positional information. A different question is whether the same proportions can be maintained when an embryo is larger or smaller. In the frog embryo, we now encounter actual cells that secrete extracellular molecules and respond to signals from other cells.

The **Spemann–Mangold organizer** is a region near the dorsal lip of the early amphibian gastrula. In the classical transplantation experiment, tissue from this region was moved to the ventral side of a host embryo. The graft contributed to an additional body axis and induced neighboring host cells to adopt new fates. The original experiments were performed with salamander embryos and published by Hans Spemann and Hilde Mangold in 1924; the frog *Xenopus laevis* is used in the modern experiments discussed here.[10, 11]

Two kinds of cutting experiment must be distinguished. A cut that leaves dorsal organizer-forming tissue in both pieces can produce two smaller, proportioned embryos. When an embryo is instead divided into dorsal and ventral halves, the dorsal half can develop into a smaller, proportioned embryo, whereas the

ventral half does not form a complete body axis. "Half an embryo" refers to the starting material; it does not mean that the resulting tadpole has exactly half the normal length.[11]

In *Xenopus*, the organizer secretes molecules including Chordin, which binds BMP ligands and inhibits their signaling while bound. Ben-Zvi and colleagues' 2008 work examined a **shuttling** mechanism: Chordin-bound BMP ligands are transported, and cleavage of Chordin releases them. Admp is another BMP ligand, produced dorsally, whose production is repressed by BMP signaling. The combination of transport and feedback provides a model for how the BMP activity profile can adjust to embryo size.[8] We will use this experimentally motivated model to introduce a general mathematical mechanism; its simplified equations are not a complete description of every molecular interaction in the embryo.

5.2 Mechanism for achieving scaling in embryos: Expansion-repression feedback

In this section, we study a mathematical model proposed by Danny Ben-Zvi and Naama Barkai that can achieve scaling of morphogen gradients ("Scaling of morphogen gradients by an expansion-repression integral feedback control", *PNAS* (2010)).[9] The model relies on a mechanism that the authors called **expansion-repression feedback**. The 2010 model extracts a general feedback principle from earlier work, including the 2008 frog study. Its abstract variables, *Morph* and *Exp*, need not each represent a single molecule in the full Xenopus network.

5.2.1 A fixed diffusion length does not automatically scale with embryo size

Before delving into the expansion-repression feedback that Ben-Zvi and Barkai proposed (motivated in part by their earlier work on frog embryos), let us consider a simpler picture: A single morphogen, *Morph*, diffusing and degrading in a field of cells. For simplicity, we will consider diffusion along a line (one-dimensional diffusion) with the cells of the embryo sitting on the line of length L, one next to the other. Suppose that at the **proximal end** ($x = 0$), there is a source of *Morph* that produces *Morph* at some rate ρ. At the **distal end** ($x = L$), we can impose at least two types of boundary conditions:

- **Dirichlet condition**: We specify $M(L, t)$ for each time t.
- **Neumann condition**: We specify the **outward normal derivative**, $\partial M/\partial n$ at $x = L$ for each time t. The outward normal derivative points away from the domain of the reaction-diffusion equation. Thus, at $x = L$, $\partial M/\partial n = \partial M/\partial x$ whereas at $x = 0$, $\partial M/\partial n = -\partial M/\partial x$

The reaction-diffusion equation for describing the concentration M of *Morph* on the line is

$$\frac{\partial M}{\partial t} = D_M \frac{\partial^2 M}{\partial x^2} - G(M) \tag{54}$$

where D_M is the diffusion coefficient of *Morph*, and $G(M)$ is a function of M and is a degradation rate of *Morph* that is either zero or a positive number. A simple example of G that we have been using in the previous lectures is a **first-order degradation**: $G(M) = \alpha M$. Let us now solve Eq. 54 for a special case.

Special case: Steady-state solution of Eq. 54 when there is no degradation: Here we set $G = 0$ (no degradation) and $\partial M/\partial t = 0$ (steady-state). Then we have the following **Laplace's equation**:

$$0 = \frac{d^2 M_s}{dx^2} \tag{55}$$

For the fixed concentrations at both ends that we impose below, the steady-state solution M_s does not depend on the diffusion coefficient D_M. To obtain M_s, we must specify the boundary conditions at $x = 0$

and $x = L$. Let us suppose that the source at the proximal end ($x = 0$) secretes and "takes away" *Morph* such that $M(x = 0, t)$ is a constant value M_0 at all times (thus $M_s(0) = M_0$). Let us also assume that at the distal end ($x = L$), we have a perfect sink (i.e., $M(x = L, t) = 0$ at all times). Thus $M_s(L) = 0$. With these Dirichlet conditions, the solution of Eq. 55 is

$$M_s(x) = M_0\left(1 - \frac{x}{L}\right) \tag{56}$$

We can check that Eq. 56 is indeed a solution of Eq. 55 by noting that (1) its second-derivative with respect to x yields zero for $0 < x < L$, and that (2) $M_s(0) = M_0$ and $M_s(L) = 0$ (i.e., satisfies both boundary conditions). So we know that Eq. 56 is indeed a solution. But how do we know that it is the *only* solution of Eq. 55? In one dimension, integrating $d^2M_s/dx^2 = 0$ twice gives $M_s = Ax + B$, and the two boundary values determine A and B uniquely. A proof that also applies in higher dimensions is given in Appendix A.

Let us now return to the one-dimensional diffusion of *Morph* between $x = 0$ and $x = L$. The steady-state solution M_s, in fact, allows for **scaling**. That is, if we have two embryos - one has length L and the other with length βL - and M_L and $M_{\beta L}$ are the steady-state concentrations of the two embryos respectively, then

$$M_L(x_1) = M_0\left(1 - \frac{x_1}{L}\right) \tag{57a}$$

$$M_{\beta L}(x_2) = M_0\left(1 - \frac{x_2}{\beta L}\right) \tag{57b}$$

where x_1 is the position within the embryo of length L and x_2 is the position within the embryo of length βL. For corresponding positions, set $x_2 = \beta x_1$. Note that if we scale both x_1 and L in Eq. 57a by β, then we obtain

$$1 - \frac{\beta x_1}{\beta L} = 1 - \frac{x_2}{\beta L} \tag{58}$$

Thus we can write M_L and $M_{\beta L}$ as

$$M_L(x_1) = M_0(1 - y) \tag{59a}$$

$$M_{\beta L}(x_2) = M_0(1 - y) \tag{59b}$$

where y is the position divided by the embryo's length. Mathematically, this is what it means to have scaling. Biologically, scaling means that the morphogen concentration within an embryo is determined by the distance from the proximal end relative to the embryo's total length and not by the absolute distance from the proximal end. But this scheme is sensitive to the imposed value of M_0: a fractional change in M_0 produces the same fractional change in the whole concentration profile, and shifts positions selected by fixed concentration thresholds. Moreover, we assumed that *Morph* does not degrade in the interior and that both boundary concentrations are maintained. Thus this special example demonstrates scaling, but does not by itself explain how an embryo maintains scaling when production, degradation, or boundary conditions vary.

5.2.2 Expansion-repression feedback

We now describe a mechanism that can robustly generate scaling in embryos. As in the previous section, we consider a diffusing morphogen called *Morph*. Being a morphogen, its concentration M is sensed by cells or nuclei inside the embryo and determines which type of cells and where they are generated inside the embryo. Let us assume that a localized source secretes *Morph* and that *Morph* then diffuses along one-dimension as before. Let D_M be *Morph*'s diffusion coefficient and α_M be its first-order degradation rate constant. Then the reaction-diffusion equation for M is again given by

$$\frac{\partial M}{\partial t} = \frac{\partial}{\partial x}\left(D_M \frac{\partial M}{\partial x}\right) - \alpha_M M \tag{60}$$

where D_M is the diffusion coefficient of *Morph* and α_M has units of inverse time; the concentration lost per unit time is $\alpha_M M$ (i.e., we have set $G(M)$ in Eq. 54 to be $\alpha_M M$). The diffusion term reduces to $D_M \partial_x^2 M$ when D_M is spatially uniform. Writing it in flux form also covers spatially varying diffusivity under Fick's law. Crucial to the authors' model is that at least one of D_M and α_M depends on the expander, as we will see shortly. We impose the following boundary condition at the proximal end: there is a constant flux η_M of M at the source located at $x = 0$. At the distal end ($x = L$), we can assume a number of different boundary conditions. The authors examined different distal boundary conditions. Here we will first use a reflective boundary (i.e., $\partial M/\partial x = 0$ at $x = L$); we will explicitly identify a different boundary approximation when using it below. Moreover, the authors consider a second diffusing molecule, called the **expander** *Exp*, whose concentration either directly or indirectly affects D_M and the degradation rate through α_M. Thus D_M and α_M are functions of M and E:

$$D_M = D_M(E, M) \tag{61a}$$
$$\alpha_M = \alpha_M(E, M) \tag{61b}$$

While we do not specfy the functional forms of D_M and α_M, the expander can promote spreading through either or both of the following effects:

- As the concentration E increases, D_M increases. Thus the expander directly/indirectly **facilitates diffusion** of *Morph*.
- As E increases, α_M decreases. Thus the expander directly/indirectly **inhibits degradation** of *Morph*.

We also assume that *Morph* represses the production of *Exp*, which we characterize through a sigmoidal production-rate function as a function of the concentration M. Then the following coupled reaction-diffusion equations describe the *Morph* and *Exp* concentrations throughout the embryo:

$$\frac{\partial M}{\partial t} = \frac{\partial}{\partial x}\left(D_M \frac{\partial M}{\partial x}\right) - \alpha_M M \tag{62a}$$
$$\frac{\partial E}{\partial t} = D_E \frac{\partial^2 E}{\partial x^2} - \alpha_E E + \beta_E \frac{T_{rep}^h}{T_{rep}^h + M^h} \tag{62b}$$

Here, T_{rep} is the repression threshold, h is a Hill coefficient, D_E is diffusion constant for E, and α_E is the first-order degradation rate constant for E. The maximum production rate is β_E. We impose $-D_M \partial_x M(0,t) = \eta_M$, $\partial_x M(L,t) = 0$, and reflective boundaries for E at both ends. Eqs. 62a and 62b define the **expansion-repression feedback** system. We next analyze these equations, first using analytical estimates and next with a "numerical screening" method.

5.2.3 Analytical approximation

Analytical estimates before numerical simulations can provide mechanistic insights into the equations that numerical simulations alone may not make apparent. We can analytically approximate solutions to Eqs. 62a and 62b before numerically solving them. For a complex equations with many parameters like these, it often helps to rescale the variables, x and t, which means that we measure the length and time relative to other parameters that have dimensions of length and time respectively (thus x and t would become dimensionless). For the analytical approximation, we first assume that the expander is nearly uniform in space, and that D_M and α_M depend on E but not directly on M. The morphogen coefficients are then uniform in space. Practically, we can expose characteristic scales by dividing Eq. 62a by α_M, which has units of 1/time, to get

$$\frac{1}{\alpha_M}\frac{\partial M}{\partial t} = \frac{D_M}{\alpha_M}\frac{\partial^2 M}{\partial x^2} - M \tag{63a}$$

$$\implies \tau\frac{\partial M}{\partial t} = \lambda^2\frac{\partial^2 M}{\partial x^2} - M \tag{63b}$$

where $\tau = 1/\alpha_M$ is a **characteristic time** and $\lambda = \sqrt{D_M/\alpha_M}$ is a **characteristic length**. The variables become dimensionless only after we introduce a scaled position and time, such as $y = x/\lambda$ and $s = t/\tau$ when the coefficients are fixed. We can similarly rescale Eq. 62b but that is not yet necessary. In the analyses below, we will assume that

$$D_M = D_M(E) \qquad \text{(Independent of } M\text{)} \tag{64a}$$

$$\alpha_M = \alpha_M(E) \qquad \text{(Independent of } M\text{)} \tag{64b}$$

Let us now consider what the steady-state solution $M_s(x)$ of Eq. 63b would look like, without actually solving the equation. Having identified λ and τ as the relevant dimensions enable us to determine the main properties of $M_s(x)$. First, we note that, in steady-state (i.e., $\partial M_s/\partial t = 0$), Eq. 63b becomes

$$0 = \frac{\partial^2 M_s}{\partial y^2} - M_s \tag{65}$$

where $y = \frac{x}{\lambda_s}$ and $\lambda_s = \sqrt{D_M(E_s)/\alpha_M(E_s)}$ is spatially constant under the uniform-expander approximation. Thus,

$$M_s(x) = f(y) \tag{66}$$

where f is some function. Furthermore, by imposing that the **flux** at $x = 0$ - this is $-D_M\partial M_s/\partial x|_{x=0}$ by Fick's law for diffusion - is η_M, we have

$$\eta_M = -D_M\frac{\partial M_s}{\partial x}\bigg|_{x=0} \tag{67a}$$

$$\implies \frac{\lambda_s\eta_M}{D_M} = -\frac{\partial M_s}{\partial y}\bigg|_{y=0} \tag{67b}$$

Eq. 67b expresses the boundary condition at $x = 0$ in terms of the rescaled position variable y. At the other boundary ($x = L$), we retain the reflective condition $\partial_x M_s(L) = 0$. Define

$$\rho_s(E_s) = \frac{\lambda_s \eta_M}{D_M(E_s)}, \qquad q = \frac{L}{\lambda_s}. \tag{68}$$

The parameter ρ_s has units of concentration. The steady-state profile can therefore be written in the form

$$M_s(x) = F\left(\frac{x}{\lambda_s}; \rho_s, q\right). \tag{69}$$

The last argument matters: a finite domain brings in the dimensionless length L/λ_s. Finding a function of x/λ_s is not yet the same as finding a function of x/L that is independent of L.

So far, we have not yet demonstrated if the proposed mechanism, the so-called "expansion-repression feedback", allows for scaling. To address this issue, we need to determine whether $M_s(x)$ admits scaling as a function of the proximal-distal length L. Our starting point is to note that increasing E can increase D_M, decrease α_M, or do both - these are the effects we imposed on the model. Thus, the steady-state value of $Morph$'s diffusion length $\lambda_s = \sqrt{D_M/\alpha_M}$ increases as E increases. This means that $Morph$ will travel further away from the proximal tip ($x = 0$), and thus we say that the morphogen region **expands** in the field. But as the concentration M of the morphogen increases at a given location, the production rate of the Exp will decrease since $Morph$ represses production of Exp - This is where the term **repression** in the "expansion-repression" mechanism arises. The repressed production of Exp, in turn, slows its accumulation. If degradation exceeds production, E decreases and the $Morph$ field can contract; if expander degradation is negligible, shutting off production instead stops further expansion. At this point, without doing calculations, we can envision at least two possibilities: (1) oscillatory behaviour in which the whole cycle of expansion-contraction of the morphogen field repeats over and over, or (2) a steady-state value of E is reached. Writing Eq. 69 does not establish which possibility occurs: we assumed a steady state to obtain it. Its existence and dynamical stability must still be checked for the chosen parameters.

To analytically extract the behaviour of $M_s(x)$, Let us assume that Exp diffuses very fast and degrades very slowly. More precisely, the mixing time L^2/D_E should be short compared with the times over which production changes the expander concentration, and $1/\alpha_E$ should be long compared with the gradient-adjustment time. Then E is approximately uniform after an initial mixing transient. Define its spatial average by $\bar{E} = L^{-1}\int_0^L E(x,t)\,dx$. Integrating Eq. 62b and using the reflective expander boundaries gives

$$\frac{d\bar{E}}{dt} = \frac{\beta_E}{L}\int_0^L \frac{T_{rep}^h}{T_{rep}^h + M(x,t)^h}\,dx - \alpha_E \bar{E}. \tag{70}$$

This equation shows how expander accumulation records production across the whole field. When the morphogen range is too short, distal cells produce expander, and the accumulated expander increases the range. With sufficiently sharp repression and very small expander loss, the system can approach a state in which production is strongly suppressed throughout the field. For a monotonically decreasing morphogen profile, the last region to be repressed is near $x = L$. In this limit, the approximate feedback condition is

$$M_s(L) \simeq T_{rep}. \tag{71}$$

This is a consequence of feedback in this approximation, not a Dirichlet boundary value imposed in addition to the reflective boundary condition. With finite α_E, residual production balances expander degradation. With $\alpha_E = 0$ and finite h, the Hill production term is strictly positive at any finite M, so an exact finite steady state with zero production is not obtained. The ideal shutoff picture uses a sharp-switch limit, fast expander mixing, and a morphogen profile that adjusts before the expander accumulates appreciably. The full equations retain finite rates.

A scaling profile must have the same value at the same x/L in embryos of different lengths. In Eq. 69, this requires control of both $q = L/\lambda_s$ and the amplitude parameter ρ_s, or sufficiently weak dependence on their remaining variation. Feedback can make λ_s track L, but simply writing $\lambda_s = L/q$ does not prove that q is independent of L. We next check that dependence explicitly.

Up to this point, the argument was very abstract. Let's be concrete by doing concrete calculations. We will next consider two examples:

- First-order degradation of morphogen
- Second-order degradation of morphogen

Case 1: First-order degradation of the morphogen: The reaction-diffusion equation with a first-order degradation term, $G(M) = \alpha_M M$, is given by

$$\tau \frac{\partial M}{\partial t} = \lambda^2 \frac{\partial^2 M}{\partial x^2} - M \tag{72}$$

where $\tau = 1/\alpha_M$ is called **characteristic time** and $\lambda = \sqrt{D_M/\alpha_M}$ is called the **characteristic length** (τ is the degradation time and also the diffusion time across the length λ). Note that λ is actually a function of the expander concentration E. In steady-state, we have $\partial M/\partial t = 0$ and $\lambda = \lambda_s$, where λ_s is a spatially constant steady-state value under the uniform-expander approximation. Thus we obtain

$$0 = \frac{d^2 M}{dy^2} - M \tag{73}$$

where $y \equiv x/\lambda_s$. The boundary conditions are

$$-D_M \left.\frac{dM_s}{dx}\right|_{x=0} = \eta_M, \qquad -\left.\frac{dM_s}{dy}\right|_{y=0} = \frac{\lambda_s \eta_M}{D_M}, \tag{74a}$$

$$\left.\frac{dM_s}{dx}\right|_{x=L} = 0. \tag{74b}$$

At $x = 0$ there is constant flux (by Fick's law of diffusion); at $x = L$ there is no flux. Here η_M is the prescribed constant source flux. At fixed parameters and length, we can write $M_s(x) = f(y)$, but f also depends on ρ_s and $q = L/\lambda_s$. Let us now solve Eq. 73. We have

$$0 = \frac{d^2 f}{dy^2} - f \tag{75}$$

Plugging $f(y) = Ce^{ay}$ into Eq. 75, we find that $a^2 - 1 = 0$. Thus, the general solution of Eq. 75 is

$$f(y) = Ae^{y} + Be^{-y} \tag{76}$$

where A and B are constants that depend on the boundary conditions (Eqs. 74a–74b). On a finite interval, the growing exponential must not be discarded merely because it diverges as $x \to \infty$: infinity is outside this domain. Applying the two boundary conditions gives

$$B - A = \rho_s = \frac{\lambda_s \eta_M}{D_M}, \tag{77a}$$

$$Ae^{q} - Be^{-q} = 0, \qquad q = \frac{L}{\lambda_s}. \tag{77b}$$

Thus $A = \rho_s/(e^{2q} - 1)$ and $B = \rho_s/(1 - e^{-2q})$.

We assume here that the expander concentration is uniform throughout the embryo, and hence $\lambda_s = \lambda_s([E])$ is a constant (i.e., both $D_M(E)$ and $\alpha_M(E)$ are constants).

Combining the two exponential terms, we have

$$M_s(x) = \frac{\lambda_s \eta_M}{D_M} \frac{\cosh[(L-x)/\lambda_s]}{\sinh(L/\lambda_s)}. \tag{78}$$

Here $\cosh z = (e^z + e^{-z})/2$ and $\sinh z = (e^z - e^{-z})/2$. For $L/\lambda_s \gg 1$ and positions several λ_s away from the distal end, this reduces to the familiar approximation

$$M_s(x) \simeq \frac{\lambda_s \eta_M}{D_M} e^{-x/\lambda_s}. \tag{79}$$

The exponential approximation is not uniform at the reflective distal boundary, where we should use the exact solution.

The expander-production rate is given by the sigmoidal function

$$P_E(M) = \beta_E \frac{T_{rep}^h}{T_{rep}^h + M^h}. \tag{80}$$

When h is large, this production becomes small for $M > T_{rep}$. At $M = T_{rep}$ it is $\beta_E/2$, even as $h \to \infty$; threshold pinning is therefore an approximation to the feedback dynamics, not an exact zero of this smooth function. In the regime where Eq. 71 applies, the exact finite-domain solution gives

$$T_{rep} \simeq M_s(L) = \frac{\lambda_s \eta_M}{D_M \sinh(L/\lambda_s)}, \tag{81a}$$

$$q \sinh q \simeq \frac{\eta_M L}{D_M T_{rep}}, \qquad q = \frac{L}{\lambda_s}. \tag{81b}$$

Consequently,

$$M_s(x) \simeq T_{rep} \cosh\left[q\left(1 - \frac{x}{L}\right)\right]. \tag{82}$$

Although position now appears as x/L, the parameter q can still change with L, and D_M can also change through the feedback. Thus linear degradation with this feedback does not automatically give exact scaling. It can give approximate scaling in suitable parameter ranges; this must be tested rather than inferred from the notation x/L.

Case 2: Second-order degradation: If the degradation rate of M is second-order (i.e., $G(M) = \alpha_M M^2/M_T$ in Eq. 54, where M_T is a constant), then a useful stationary solution of the differential equation is:

$$M_s(x) = \frac{6 D_M M_T}{\alpha_M (x + \epsilon)^2} \tag{83}$$

Here ϵ is a length. Differentiating twice verifies that $D_M M_s'' = \alpha_M M_s^2/M_T$. Imposing the source flux gives

$$\eta_M = -D_M M_s'(0) = \frac{12 D_M^2 M_T}{\alpha_M \epsilon^3}, \qquad \epsilon = \left(\frac{12 D_M^2 M_T}{\alpha_M \eta_M}\right)^{1/3}. \tag{84}$$

Unlike Eq. 78, this power-law solution is not an exact solution with a reflective boundary at finite L, since its derivative there is nonzero. For this analytically tractable example, we instead use the idealization

of a profile that continues beyond the responsive field $0 \leq x \leq L$ and tends to zero at infinity, with the same constant effective coefficients. This boundary approximation must be kept separate from the closed finite-domain problem; that problem can be solved numerically.

Applying the approximate feedback condition $M_s(L) \simeq T_{rep}$ in this example gives

$$\frac{6D_M M_T}{\alpha_M} \simeq T_{rep}(L+\epsilon)^2, \qquad M_s(x) \simeq T_{rep}\left(\frac{L+\epsilon}{x+\epsilon}\right)^2. \tag{85}$$

In the limit $L \gg \epsilon$ and at positions $x \gg \epsilon$, Eq. 83 therefore becomes

$$M_L(x/L) \approx \frac{T_{rep}}{(x/L)^2} \tag{86}$$

According to Eq. 86, quadratic degradation together with the feedback and the stated boundary approximation yields a nearly perfect scaling away from the source - the steady-state concentration of morphogen depends only on the relative position x/L, and no factors of L and x appear by themselves Eq. 86. But with the quadratic degradation of $Morph$, the scaling is still not perfect. In fact, when x is comparable to ϵ (i.e., near the proximal end where $Morph$ is produced), the dependence on ϵ/L cannot be dropped, so this calculation no longer establishes scaling. The apparent divergence of Eq. 86 at $x = 0$ is likewise outside its range of validity; the unapproximated profile remains finite there.

5.3 Testing the approximation by numerical screening

We have so far used gross approximations (e.g., very fast diffusion and very slow degradation of Exp). But we would now like to know whether the scaling still holds for less drastic values of the parameters, and if so, precisely for which values for each parameter. Our job now is to iteratively assign different numbers to each parameter in the equation and check, for each set of assigned values, if system yields scaling. This procedure, called "**numerical screening**", is often used in systems biology model. The moral here is that two equations that have the same functional forms and same variables can yield qualitatively distinct behaviours if we assign different numbers to the same parameters.

For each parameter set, compare the morphogen profiles at two lengths, plotting both against x/L. A useful test follows three concentration thresholds and asks whether the corresponding positions occupy the same fractions of the two lengths. This tests *scaling with size*. Separately changing the source strength tests *robustness to production changes*; one result does not imply the other. The full equations, boundary conditions, screening criteria, and scoring procedure are retained in Appendix B.[9]

6 Three questions answered by reaction–diffusion models

The same mathematical framework has served three distinct purposes. For Turing instability, we began with a homogeneous steady state and asked whether reactions and diffusion could amplify a nonuniform perturbation. For Bicoid, a localized maternal source already supplied a spatial asymmetry, and we asked how a gradient could specify position with adequate precision. For scaling, we asked how feedback could adjust a gradient when the size of the field changed.

These questions require different calculations. Linear stability tests the initial growth of perturbations. The gradient slope and sensing noise determine a positional error. Comparing profiles at the same fractional position tests scaling. A pattern can be stable without scaling, and a scaling concentration profile can still be read noisily. Keeping these questions separate lets us connect each model prediction to the appropriate measurement.

A Laplace equation: maximum principle and uniqueness

Theorem 1 (No local extrema): Let D be a bounded, connected open domain, and let M be twice continuously differentiable in D and continuous on its closure, with $\nabla^2 M = 0$. Its maximum and minimum over the closure are attained on the boundary; if a global maximum or minimum is attained inside D, then M is constant. A nonconstant harmonic function also has no interior local maximum or minimum.
Proof: We will use another fact about Laplace's equation without proof here - A solution M to a Laplace equation $\nabla^2 M = 0$ has the property that its value $M(\vec{r}_0)$ at point $\vec{r}_0$ inside the domain D is the average of all the values of M on the surface of any sphere whose center is at $\vec{r}_0$ and fits inside D. In 1-, 2-, and 3-dimensions, this means

$$M(x_0) = \frac{M(x_0 - a) + M(x_0 + a)}{2} \qquad \text{(1 dimension)}, \tag{87a}$$

$$M(\vec{r}_0) = \frac{1}{2\pi R} \oint_{\text{circle}} M(\vec{r})\, ds \qquad \text{(2 dimensions)}, \tag{87b}$$

$$M(\vec{r}_0) = \frac{1}{4\pi R^2} \int_{\text{sphere}} M(\vec{r})\, dS \qquad \text{(3 dimensions)}. \tag{87c}$$

In one dimension, $x_0 - a$ and $x_0 + a$ are points in interval D. In two and three dimensions, the circle or sphere has radius R, is centered at $\vec{r}_0$, and its enclosed disk or ball lies inside D.

For the maximum over the whole domain, suppose M attains this value at an interior point $\vec{r}_0$. On every sufficiently small sphere around $\vec{r}_0$, all values are at most $M(\vec{r}_0)$. The mean-value property and continuity force every value on each such sphere to equal $M(\vec{r}_0)$: if one value were strictly smaller, a neighborhood on that sphere would be smaller and the average would be smaller. Thus the set of points attaining the maximum is open as well as closed in D. Since D is connected, M is constant throughout D. Applying the same argument to $-M$ gives the statement for a minimum. For a merely local maximum, the same argument first shows that M is constant in a neighborhood. Harmonic functions are real analytic (a standard result not proved here), so constancy on this open neighborhood extends throughout a connected domain. Thus theorem 1 holds.

Theorem 2 (Uniqueness theorem): Under the bounded-domain and regularity assumptions of theorem 1, a solution to the Laplace's equation with prescribed Dirichlet boundary values is unique if it exists. Thus we can call it *the* solution.
Proof: We use theorem 1 to prove this. Suppose, for contradiction, that we have two distinct solutions, M_1 and M_2 in the same domain D and with the same Dirichlet boundary conditions. Then since the Laplace's equation is a linear equation, we must have $\nabla^2(M_1 - M_2) = \nabla^2 M_1 - \nabla^2 M_2 = 0$. Thus $M_1 - M_2$ is a solution inside the domain D as well. On the boundary of D, $M_1 - M_2$ is zero since M_1 and M_2 take the same values there. By theorem 1, we must then have $M_1 - M_2 = 0$ everywhere inside D since, for otherwise, $M_1 - M_2$ would have a local extremum inside D. Thus $M_1 = M_2$ in the end. Thus once we find a solution M_1, we have found all the solution there is for the Laplace's equation on D with a fixed Dirichlet boundary condition. Thus theorem 2 holds.

B Numerical screening

We have so far used gross approximations (e.g., very fast diffusion and very slow degradation of Exp). But we would now like to know whether the scaling still holds for less drastic values of the parameters, and if so, precisely for which values for each parameter. In the previous section, we obtained the functional forms of the equations that describe the diffusion and degradations of morphogen and expander. Our job now is to iteratively assign different numbers to each parameter in the equation and check, for each set of assigned values, if system yields scaling; robustness to parameter changes is tested separately. This procedure, called "**numerical screening**", is often used in systems biology model. The moral here is that two equations that have the same functional forms and same variables can yield qualitatively distinct behaviours if we assign different numbers to the same parameters. The set of equations that the authors numerically screened is

$$\frac{\partial M}{\partial t} = D_M \frac{\partial^2 M}{\partial x^2} - (1 + E/E_0)^{p_1} \alpha_1 M - (1 + E/E_0)^{p_1} \alpha_2 M^2 \tag{88a}$$

$$\frac{\partial E}{\partial t} = D_E \frac{\partial^2 E}{\partial x^2} - \alpha_E E + \beta_E \frac{(M/T_{p_2})^{hp_2}}{1 + (M/T_{p_2})^h} \tag{88b}$$

Here E_0 is a fixed concentration scale, so $1 + E/E_0$ is dimensionless. Equivalently, one can measure E in units of E_0 and write the original factor $1 + E$. We use the following boundary conditions

$$-D_M \frac{\partial M}{\partial x}\bigg|_{x=0} = \eta, \qquad D_M \frac{\partial M}{\partial x}\bigg|_{x=L} = 0, \tag{89a}$$

$$D_E \frac{\partial E}{\partial x}\bigg|_{x=0} = 0, \qquad D_E \frac{\partial E}{\partial x}\bigg|_{x=L} = 0. \tag{89b}$$

These are Neumann conditions: constant flux of $Morph$ at the proximal end; no flux of $Morph$ at the distal end; and no flux of Exp at either end (reflective conditions).

In Eqs. 88a and 88b, we assign different numbers to the following constants and check if a particular set of values yield scaling:

- D_M: Diffusion coefficient for $Morph$.
- D_E: Diffusion coefficient for Exp.
- α_1: linear degradation term for $Morph$.
- α_2: quadratic degradation term for $Morph$.
- α_E: linear degradation term for Exp.
- η: Flux of $Morph$ at the proximal end ($x = 0$).
- β_E: maximum production rate of Exp.
- p_1: This is either -1 (Exp **decreases** degradation of $Morph$) or +1 (Exp **increases** degradation of $Morph$).
- p_2: This is either 0 ($Morph$ represses Exp) or +1 ($Morph$ induces Exp).
- T_{p_2}: T_0 is repression threshold for Exp and T_1 is induction threshold for Exp.
- h: Hill coefficient.

Procedure for numerical screening of Eqs. 88a and 88b: Ben-zvi and Barkai assigned approximately 400,000 different sets of numbers to the above parameters and, for each of these parameter sets, used MATLAB to numerically solve Eqs. 88a and 88b. MATLAB has built-in PDE solvers. "Numerically solving" means using one of these solvers, the authors obtained graphs of M and E versus x/L. They then compared the morphogen concentration profiles for (1) normal length (L=100μ m) and (2) double length (L=200μ m). If there was a strong overlap between the two morphogen profiles (both functions of x/L), then they concluded that the particular set of parameters yielded scaling. But having scaling is not sufficient if the values of M and E are biologically non-sensical. To make biological sense, we retain the following operational criteria for the screening procedure:

Conditions for M to be a biologically valid concentration-profile:

1. M must be maximum at $x = 0$ (where the source is) and minimum at $x = L$ (distal end) with its maximum being larger than 10 times its minimum.
2. Concentration at $x = 0.5L$ or $x = 0.75L$ must be larger than $10^{-6}\mu$ M.
3. The ratio between the maximum of M and the value of M at $x = 0.25L$ is larger than 1 and less than 100 (i.e., rule out extremely sharp concentration profiles).
4. M and E both reach their steady-state. To overcome numerical fluctuations in MATLAB's solver, we first pick two time points - $t_{1/2}$ and $2t_{1/2}$ - where $2t_{1/2}$ is the total time that we impose on simulating Eqs. 88a and 88b. We then measure the average values of M and E during the time interval $[t_{1/2}, 2t_{1/2}]$ at each of the following five positions: $x = 0$, $x = 0.25L$, $x = 0.5L$, $x = 0.75L$, and L. If M's and E's averages deviate less than 1% from their values at time $2t_{1/2}$ at all five positions, then the system passes this numerical steady-state check. This finite-time check is a practical screening criterion, not a mathematical proof of convergence.

Using the four criteria above, the authors found that only 7% of the 400,000 parameter sets were biologically valid. The authors then scored each of the 7% of the parameter sets based on how well they scaled with L.

To quantify how well the two graphs overlapped, the authors assigned a score to each parameter set using the following scoring metric:

Scoring metric:

1. For L_1=100 μm: Find the steady-state value of M at three positions: $y_1 = x_1/L_1 = 0.25$, $y_2 = x_2/L_1 = 0.5$, and $y_3 = x_3/L_1 = 0.75$. Let these three values be M_1, M_2, and M_3.
2. For L_2=200 μm: Find positions $y_1^* = z_1/L_2$, $y_2^* = z_2/L_2$, and $y_3^* = z_3/L_2$ where the steady-state value of M is M_1, M_2, and M_3.
3. Compute $\sigma = \frac{1}{3}\sum_{i=1}^{3} |\delta_i|$, where $\delta_i = y_i - y_i^*$

Note that $\sigma = 0$ would be a **perfect scaling at the three sampled thresholds** whereas unchanged absolute threshold positions ($z_i = x_i$) when the length doubles give $\sigma = 0.25$. The latter is a useful non-scaling reference case, not the score of every possible non-scaling profile. Furthermore, note that if one of the three values - M_1, M_2, and M_3 - was not reached when the embryo doubled in length, then scaling clearly does not occur. The authors then selected parameter sets with $\sigma < 0.1$ as positive for scaling under this criterion. A separate robustness test halves and doubles the source flux and checks that the average displacement of these threshold positions, expressed as a fraction of the field length, is less than 0.1.[9]

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